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Research Article

Finite Element Analysis of Local pH Variations in Electrolysis with Porous Electrodes, Considering Water Self-Ionization

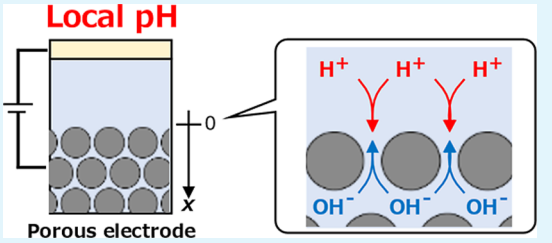
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ABSTRACT: A finite element model was developed to simulate ion fluxes and local pH changes within and around porous electrodes during the H_2 evolution reaction (HER) in acidic electrolytes. This model is particularly characterized by its ability to simulate scenarios in which the local pH inside and near the cathode exceeds 7, even under bulk acidic conditions (e.g., pH = 1), by considering the self-ionization of water. Steady-state calculations using meshes with an appropriate spatial distribution inside and near the cathode revealed that the relationship between the local pH and the double-layer potential at the interface between the porous catalyst layer and the electrolyte domains changed notably when the local pH exceeded the threshold of 7. By comparing the fluxes of H^+ and OH^- ions at the interface and using the thickness of the catalyst layer as a variable, we determined that the presence of H^+ ions within the pores or the supply of OH^- ions from the pores to the interface was responsible for the characteristic change in the local pH observed for the porous electrode. The porous electrode model constructed in this study can potentially serve as a basis that can be extended to a wide range of electrolysis systems, including not only the HER, but also the reduction of CO_2, H_2O_2, and O_2, and even oxidation reactions such as the O_2 evolution reaction.



KEYWORDS: *hydrogen evolution reaction, porous electrode, local pH, finite element method, Nernst−Planck equation*

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1. INTRODUCTION

Electrolysis has attracted intensive interest as a means to achieve a zero-emission society.[1,2] In electrolysis technologies, porous electrodes, including gas diffusion electrodes, are commonly used to achieve practically high current densities.[3−5] Under conditions of high current density, the local pH near the reaction interface can substantially increase or decrease as reduction or oxidation reactions occur, respectively. The change in local pH within the porous electrode can be further intensified by the restricted diffusion of H^+ or OH^- in the pores. A large change in the local pH increases the concentration overvoltage, inevitably leading to an increase in the voltage required for electrolysis. In addition, the local pH can influence various factors, such as reaction selectivity, wettability, and material stability. Therefore, quantitatively estimating the local pH during the reactions is critical for the to rational design of electrolysis systems.

Several experimental methods have been used to estimate the local pH.[6−10] For example, Yang et al. used in situ surface-enhanced infrared absorption spectroscopy to analyze the local pH by measuring the phosphate buffer equilibrium near the surface of copper electrodes.[7] Welch et al. combined a pH indicator with confocal microscopy to study the dependence of the local pH on the current density.[8] The local pH was measured inside and outside a porous electrode, as far as ±30 μm in the vertical direction from the electrode surface. However, because such local pH measurement methods necessitate a specialized setup, they are difficult to apply to practical reactors equipped with porous electrodes.

In light of these experimental challenges, quantitatively estimating the local pH within and near a porous electrode through numerical simulation, using a reaction-diffusion model is an effective approach.[11−15] For instance, Weng et al. developed a gas diffusion electrode model that focuses on the degree of flooding within a porous electrode and calculated the pH distribution inside the electrode during alkaline CO_2 electrolysis.[12] Chen et al. used simulations to estimate the extent to which a gas diffusion electrode improves the limiting current density for CO and CO_2 electrolysis and presented the local pH profile within the gas diffusion electrode.[13] In these previous numerical studies, the bulk pH was assumed to be greater than 7. Notably, however, even in an acidic bulk electrolyte, the local pH near the cathode can increase to above 7 during electrolysis, potentially reaching 14 depending on the current density.[16−18] Thus, systems with local pH values varying beyond 7 have been constructed, providing valuable information regarding experimental results and condi-

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tions.[16,19,20] For instance, by conducting numerical simulations, Pande et al. confirmed the validity of experimental results for time-resolved pH profiles crossing pH 7.[19] Xie et al. showed the appropriate experimental conditions for a CO_2 reduction reaction in acidic media, considering the local pH in the diffusion layer (DL) and a porous electrode under an applied current density of 500 mA cm^{-2}.[16] Zhang et al. analyzed the carbonate equilibrium during the CO_2 reduction reaction on Ag nanoparticles in mild acidic electrolytes by simulating the pH values near the catalyst surface.[20] The authors of these previous studies successfully estimated the variation of local pH and provided useful insights. However, these studies did not adequately address the specific behavior of porous electrodes. Porous electrodes have several characteristic parameters, including the size of the catalyst particles and the thickness of the catalyst layer, which make discussions of local pH variations difficult. Thus, in the present study, we focused on addressing the subsequent critical challenge: developing a model capable of comprehensively analyzing pH variations and ion fluxes within and near porous electrodes.

2. EXPERIMENTAL METHODS & RESULTS

In the present study, a simulation model was designed to reproduce the H_2 evolution reaction (HER) occurring on a Pt electrode in a 0.1 M $HClO_4$ solution. In this electrochemical system, the range of local pH variation at the electrode surface is strongly influenced by the presence or absence of alkali metal cations, leading to notable differences in the shape of the current density (j) vs the applied potential (U) curve (Figure 1).[17,21,22] The j vs U curves in Figure 1

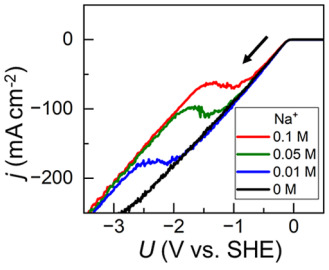


Figure 1. Linear sweep voltammograms in 0.1 M $HClO_4$ with various concentrations of $NaClO_4$. A Pt wire was used as the working electrode. The potential sweep rate was set at 10 mV s^{-1}. The black, blue, green, and red curves represent Na^+ concentrations of 0, 0.01, 0.05, and 0.1 M, respectively.

are similar to those reported in a previous paper, though obtained with different electrolyte compositions.[17] In a 0.1 M $HClO_4$ solution (black line), the cathodic current associated with the HER starts to flow at approximately −0.05 V vs standard hydrogen electrode and its absolute value increases as the potential shifts negatively. In a 0.1 M $HClO_4$ + 0.1 M $NaClO_4$ solution (red line), the cathodic current associated with the HER also begins to flow from approximately −0.05 V. However, a current plateau is observed in the potential range from −0.7 to −1.5 V, unlike the case without alkali-metal cations (Na^+). At−1.5 V, the absolute value of the reduction current density begins to increase again. The j vs U characteristics are explained as follows. The H^+ reduction reaction (eq 1) in an acidic solution proceeds at 0.0 V.

$$2H^+ + 2e^- \rightarrow H_2(g) \quad (1)$$

When the potential reaches approximately −0.7 V, the current density becomes independent of the potential because the supply of H^+ to the electrode surface becomes the rate-limiting factor, resulting in the appearance of a current plateau. The water reduction reaction (eq 2) proceeds below −1.5 V under locally alkaline conditions at the electrode surface.

$$2H_2O + 2e^- \rightarrow H_2(g) + 2OH^- \quad (2)$$

The behavior described above is influenced by the concentration of alkali-metal cations. More specifically, as the concentration of Na^+ increases, the absolute value of the current density, $|j|$, in the current plateau region decreases (blue, green, and red lines). A similar behavior is observed when using H_2SO_4 solutions containing K_2SO_4 or Na_2SO_4 or when using other metal electrodes (e.g., an Au wire electrode as shown in Figure S1) in place of Pt.[21−24] The effect of alkali-metal cations on the current plateau is universally observed in other reaction systems such as the CO_2 reduction reaction,[25] H_2O_2 reduction reaction,[26] O_2 reduction reaction,[27] and even oxidation reactions such as the O_2 evolution reaction.[28] The aforementioned observations indicate that the chemistry of the electrode reaction is not the determining factor for the development of the current plateau. Rather, the concentration of alkali-metal cations, the current density, and the surface pH are important. In the following section, first, the experimental results obtained with a planar electrode (Figure 1) are reproduced by simulation using the finite element method, which enables an analysis of the local pH fluctuation behavior near the planar electrode. Subsequently, the established protocol was applied to estimate the local pH fluctuation inside a porous electrode, which is the primary focus of this study. Notably, since the phenomenon of interest was observed regardless of the kind of alkali metal ion involved as described above, it is not necessarily required to consider the kinetic effect of alkali metal cations.

3. CALCULATION METHODS

3.1. Geometry of the Model. A one-dimensional porous electrode model (hereafter referred to as the porous model, Figure 2) was developed to conduct numerical simulations of local pH variations. The model contained bulk electrolyte, a DL, and a porous

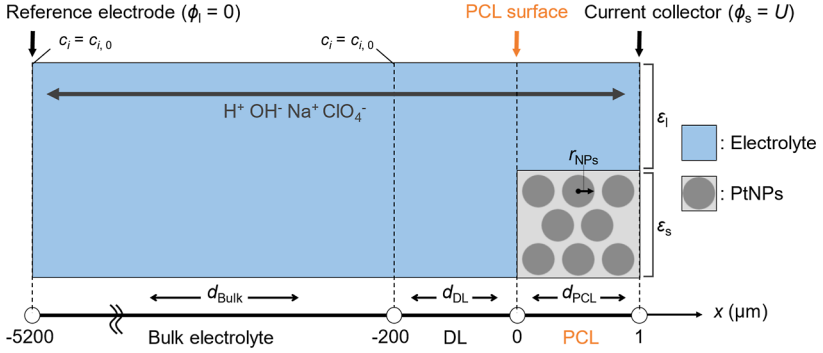


Figure 2. Schematic of one-dimensional porous electrode model (porous model) consisting of bulk electrolyte, a DL, and a PCL. The radius of PtNPs (r_{NPs}) is not to scale. The electrolyte is 0.1 M $HClO_4$ + (0, 0.01, or 0.1) M $NaClO_4$.

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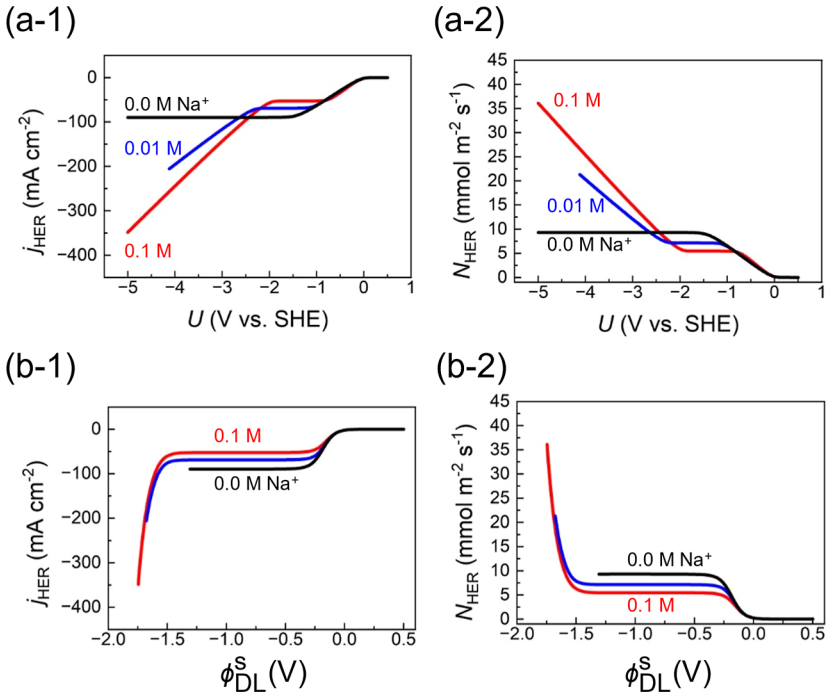


Figure 3. Calculated results for (a-1) j_{HER} vs U, (a-2) N_{HER} vs U, (b-1) j_{HER} vs ϕ^s_{DL}, and (b-2) N_{HER} vs ϕ^s_{DL} curves, as obtained using porous model. The black, blue, and red curves represent Na^+ concentrations of 0.0, 0.01, and 0.1 M, respectively.

catalyst layer (PCL). The electrolyte was 0.1 M $HClO_4$ + (0, 0.01 or 0.1) M $NaClO_4$. The electrolyte contained four ionic species: H^+, OH^-, Nah^+, and ClO_4^-. To consider the potential distribution between reference electrode and working electrode, the thickness of the bulk electrolyte (d_{bulk}) was set to 5.0 mm and the concentration of ionic species were fixed at the initial concentration ($c_{i,0}$ at the both of the bulk electrolyte boundary). That of the DL (d_{DL}) was set to a fixed value of 200 μm such that the calculated current density in the plateau region was in good agreement with the experimental value (approximately 50 mA cm^{-2}), as shown in Figure S2. The thickness of the PCL (d_{PCL}) in the porous model was set to 1 μm unless otherwise specified. The detailed calculation conditions, including the boundary conditions, are provided in the Supporting Information. To obtain fundamental understandings for the porous model, we also developed a one-dimensional planar electrode model (hereafter referred to as the planar model, Figure S3) by removing the PCL from the porous model. The cross-sectional surface area, A, was set to 0.5 cm^2 in both models. The PCL was assumed to consist of spherical Pt nanoparticles (PtNPs) with a 100 nm diameter unless otherwise specified. A macro-homogeneous porous electrode model was used to depict the distribution of particles within the PCL.[12] This model offers various insights through spatially averaged governing equations applied to simplified domains.[29] The PtNPs are virtually connected and conductive, allowing electrons to flow even though the PtNPs do not appear to be connected in Figure 2. We additionally assumed that the PCL is composed solely of solid and liquid phases, implying that any gas generated within the PCL is instantly vented from the system. The interface between the PCL and the DL is defined hereafter as the PCL surface, and the pH values at the PCL surface are specifically referred to as pH^s values.

3.2. Mathematical Description. The flux of chemical species i, N_i, is described by the sum of diffusion ($N_{i,diff}$) and electromigration ($N_{i,mig}$) terms (i.e., the Nernst−Planck equation)

$$N_i = N_{i,diff} + N_{i,mig} = -D_i \nabla c_i - z_i u_{m,i} F c_i \nabla \phi \quad (3)$$

where D_i, c_i, z_i, $u_{m,i}$, F, and ϕ are the diffusion constant, concentration, charge number, mobility of chemical species i, Faraday's constant (=96485 C mol^{-1}), and potential, respectively. The $u_{m,i}$ value was calculated on the basis of the Nernst−Einstein relation ($u_{m,i} = D_i/RT$, where R and T represent the gas constant (=8.3145 J mol^{-1} K^{-1}) and temperature, respectively). Here, convective fluxes were neglected because the focus of this paper is the flux of chemical species due to (electro)chemical reactions rather than the fluid flow. The equilibrium of water self-ionization and the electroneutrality are described by the "water-based with electroneutrality" model[30]

$$c_H c_{OH} = K_w \quad (4)$$

$$\sum_i z_i c_i + c_H - c_{OH} = 0 \quad (5)$$

$$c_H = -\frac{\sum_i z_i c_i}{2} + \sqrt{\frac{(\sum_i z_i c_i)^2}{4} + K_w} \quad (6)$$

where K_w (=1 × 10^{-14} mol^2 L^{-2}) is the equilibrium constant for self-ionization of water. Here, we assumed that electroneutrality was satisfied in all of the domains based on eq 5 (i.e., the total concentration of cations is the same as that of anions at each position x (Figure S4a,b)). The effective diffusion constant inside the porous electrode, $D_{i,eff}$, and the effective electrical conductivity in the solid phase, $\sigma_{s,eff}$, are described by the Bruggeman correction[15,31−34]

$$D_{i,eff} = \varepsilon_l^{1.5} D_i \quad (7)$$

$$\sigma_{s,eff} = \varepsilon_s^{1.5} \sigma_s \quad (8)$$

where ε_l, ε_s, and σ_s are the electrolyte volume fraction, the solid volume fraction in the PCL ($\varepsilon_l + \varepsilon_s = 1$), and the electrical conductivity of the nonporous composite electrode, respectively. Because the experimentally estimated value of the effective diffusion constants depends on the experimental conditions, the Bruggeman correction might not be adequate. Therefore, the effect of the change in the constants on the simulation was investigated by varying the multiplier of ε in eq 7. As shown in Figure S5, a variation in the multiplier of ε from 1.5 (assuming the Bruggeman correlation) to 1.2 or 1.8 did not substantially affect the outcomes of the study, indicating that the value of ε is not critical. Thus, all simulations in the following sections are based on the Bruggeman correction.

The specific active surface area of the porous electrode, A_v, was calculated as[12]

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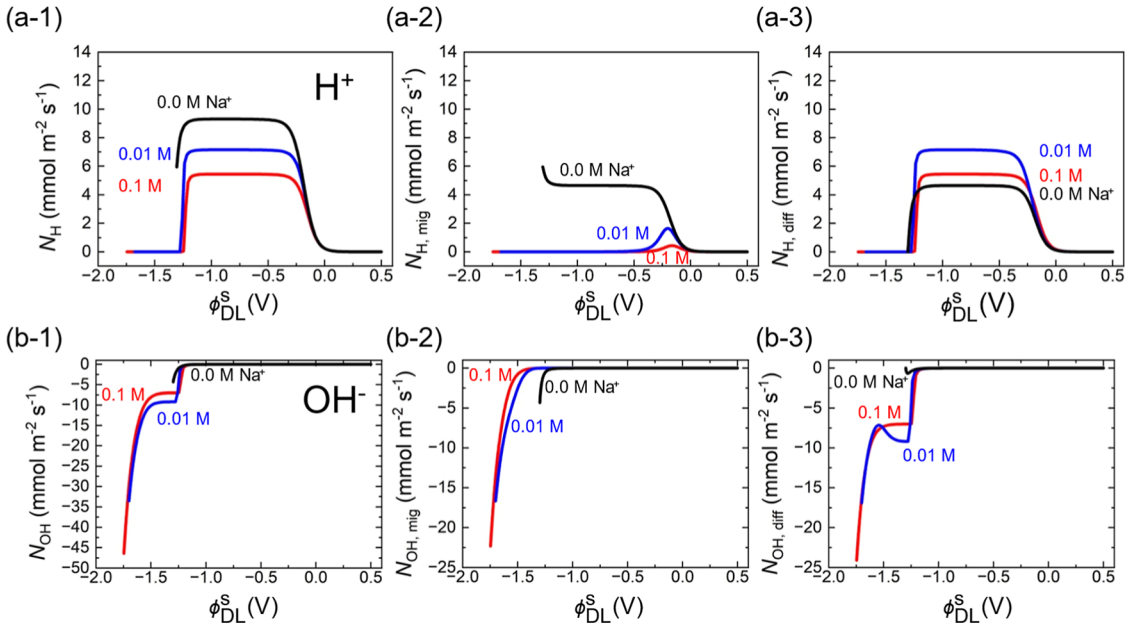


Figure 4. Fluxes of H^+ and OH^-, as obtained using porous model. (a-1) Sum of electromigration and diffusion, (a-2) electromigration, and (a-3) diffusion fluxes of H^+. (b-1) Sum of electromigration and diffusion, (b-2) electromigration, and (b-3) diffusion fluxes of OH^-. The black, blue, and red curves represent Na^+ concentrations of 0.0, 0.01, and 0.1 M, respectively.

$$A_v = \frac{3(1-\varepsilon_s)}{r_{NPs}} \tag{9}$$

where r_{NPs} is the radius of the PtNPs. The double-layer potential, ϕ_{DL}, in the PCL, is described as

$$\phi_{DL} = \phi_s - \phi_l \tag{10}$$

where ϕ_s and ϕ_l are the potentials of the solid phase (PtNPs) and that of the liquid phase (electrolyte), respectively. The current density due to the HER, j_{HER}, is given by

$$j_{HER} = \int_{PCL} A_v j_{HER,loc}\, dx \tag{11}$$

$$j_{HER,loc} = -j_{HER,0}\left(\frac{c_H}{c_{ref}}\right)\exp\left(-\frac{\alpha_{HER} n_e F}{RT}\eta_{HER}\right) \tag{12}$$

$$\eta_{HER} = \phi_{DL} - E_{HER,0} \tag{13}$$

where $j_{HER,loc}$, $j_{HER,0}$, c_{ref} α_{HER}, n_e, η_{HER}, and $E_{HER,0}$ are the local current density, exchange current density, reference concentration (1 M), transfer coefficient, transfer number of electron, overpotential, and standard electrode potential for the HER, respectively. In this model, both proton reduction (eq 1) and water reduction (eq 2) are approximately considered by assuming that the equilibrium of water (eq 4) is maintained. However, it should be noted that when the current density is very high that the rate of OH^- generation by the electrode reactions exceeds the rate of OH^- consumption by the neutralization reaction, this approximate treatment becomes inappropriate.

3.3. Calculation Method. The above equations were solved using COMSOL Multiphysics 6.0, with a MUMPS general solver. The maximum mesh size was set to 0.31 mm for the bulk electrolyte domain. To accurately calculate the pH value near the PCL surface, we adopted a variable mesh width (defined as distribution mesh in COMSOL Multiphysics) for the DL and the PCL. The DL uses a progressively finer mesh as it approaches the PCL surface from the electrolyte bulk side, whereas the PCL features a coarser mesh moving away from the PCL surface. The number of mesh elements was set to 350 and 120 for the DL and PCL, respectively. The ratio of the last mesh cell size to the first mesh cell size was set to 3×10^{-6} and 3×10^{-4} to specify the distribution for the DL and PCL, respectively. The relative tolerance was set to 1×10^{-8} or 1×10^{-7} for the porous model and planaer model, respectively. When a relatively larger relative tolerance[13,16] and/or coarser mesh near the PCL surface was used, the accuracy of the flux calculation was substantially reduced even though the j_{HER} vs U plot shows good agreement with the experimental results under the smaller relative tolerance condition (Figure S6). The model input parameters are listed in Table S1. The potential range for simulating j_{HER} under steady-state conditions was varied from +0.5 to −5.0 V in steps of 20 mV.

4. NUMERICAL SIMULATION RESULTS

4.1. Simulation of Current Density and Ion Flux for Porous Model. A simulation was conducted to study the effects of the presence of alkali metal-cations on the HER. The j_{HER} vs U curves calculated using the porous model are shown in Figure 3a-1. Note that the units for current density and flux are mA cm^{-2} and mmol m^{-2} s^{-1}, respectively. In good agreement with the experimental results in Figure 1, $|j_{HER}|$ in the plateau region decreases with increasing concentration of cations. Thus, in this section, the effect of cations on the HER is discussed on the basis of the porous model.

Notably, in the simulations, the current plateau extends to −5.0 V in the absence of Na^+ (black line). This observation is attributed to the fact that the stirring effect caused by H_2 bubbles, which invariably occurs in the experiment, is not considered in the simulation.[35] In the following, ϕ_{DL} at the PCL surface is represented as ϕ^s_{DL}. For the j_{HER} vs ϕ^s_{DL} curve, we observe that the reduction current shows a two-step increase at approximately $\phi^s_{DL} = -0.01$ and −1.44 V (Figure 3b-1). These values correspond to a reduction of H^+ (eq 1) and water (eq 2), respectively.

Next, the fluxes of H^+ and OH^- (N_H and N_{OH}, respectively) at the PCL surface were calculated using eq 3. The concentration of OH^- is represented through the concentration of H^+ as indicated in eq 4. As evident in Figure 3, j_{HER} is proportional to N_{HER} ($=-j_{HER}/n_eF$), which is defined as

$$N_{HER} = N_H - N_{OH} \tag{14}$$

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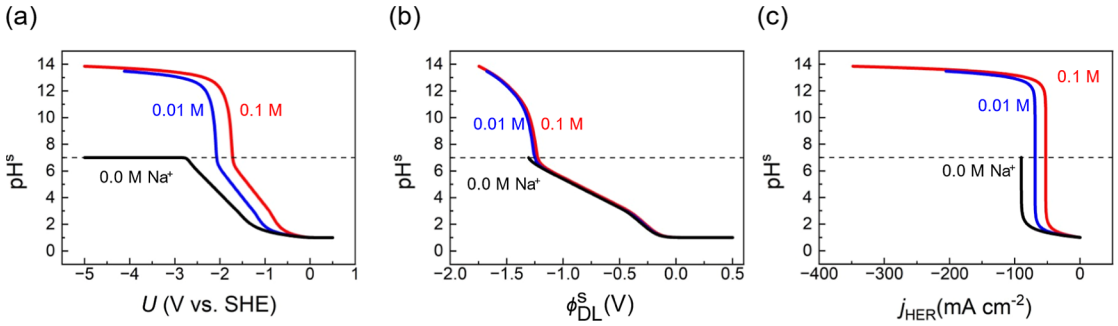


Figure 5. pH^s plotted against (a) U, (b) ϕ^s_{DL}, and (c) j_{HER}, as calculated using porous model. The black, blue, and red curves represent Na^+ concentrations of 0.0, 0.01, and 0.1 M, respectively.

here, the direction of flux from the bulk electrolyte toward the PCL surface is defined as positive, consistent with the direction of the x-axis. Note that the minus sign is necessary for N_{OH} because H^+ and OH^- ions invariably move in opposing directions. The fundamental trends of these curves are observed even when the planar model is applied (Figure S7).

As shown in Figure 1, the experimental results demonstrate (1) that $|j|$ in the plateau region decreases with increasing Na^+ concentration, and (2) that $|j|$ increases again at more negative potentials in the presence of Na^+. To clarify this behavior, we analyzed N_H and N_{OH} on the basis of the Nernst−Planck equation (eq 3). The calculated N_H and N_{OH} plotted against ϕ^s_{DL} are shown in Figure 4a-1,b-1, respectively. In the region where the current plateau appears (ϕ^s_{DL}: −0.43 to −0.96 V), the maximum value of N_H decreases with increasing Na^+ concentration (Figure 4a-1). This observed change correlates with a decrease in the plateau current as the Na^+ concentration increases. In the region where $\phi^s_{DL} < -1.12$ V, N_H decreases because the concentration of H^+ near the PCL surface becomes as low as 1×10^{-7} M (Figures 5b and S4c). Simultaneously, the absolute value of N_{OH} begins to increase (Figure 4b-1) because the concentration of OH^- increases as a result of water dissociation. In the region of $\phi^s_{DL} < -1.44$ V the OH^- starts to form by the water reduction reaction, leading to an increase in the absolute value of N_{OH}.

To demonstrate that the behavior of pH^s in the porous model resembles that in the planar model even though the models fundamentally differ, we conducted a further analysis of N_H and N_{OH} by distinguishing between the contributions of diffusion ($N_{H,diff}$ and $N_{OH,diff}$) and electromigration ($N_{H,mig}$ and $N_{OH,mig}$). Note that the difference between the porous and planar models will be discussed in Section 4.3. $N_{H,mig}$ decreases substantially at $\phi < -0.01$ V with increasing bulk Na^+ concentration (Figure 4a-2). In the potential region where the current plateau appears (ϕ^s_{DL}: −0.43 to −1.44 V), the rate of the HER is limited by the rate at whih H^+ is supplied to the PCL surface. Therefore, because of the decrease in $N_{H,mig}$ associated with the increase in the Na^+ concentration, $|j_{HER}|$ decreases in the current plateau region, in a good agreement with previously reported results.[36] $N_{H,diff}$ is larger when Na^+ is present than when it is absent (Figure 4a-3). However, as shown in Figure 4a-1, this increase in $N_{H,diff}$ does not contribute to an increase in N_H ($=N_{H,diff} + N_{H,mig}$) because the decrease in $N_{H,mig}$ is substantially greater.

As shown in Figure 3b-1, in the absence of Na^+, the current plateau is observed within a ϕ^s_{DL} range from −0.43 to −1.30 V (black curve). To gain further insight into the current plateau, $N_{OH,mig}$ is analyzed. Here, as shown in Figure 4b-2, the absolute value of $N_{OH,mig}$ increases at $\phi^s_{DL} = -1.20$ V (black curve) as ϕ^s_{DL} decreases; that is, the negative-going flux of OH^- ions resulting from water dissociation increases. However, given the constraints imposed by the water self-ionization, the concentrations of H^+ and OH^- are maintained at approximately 1×10^{-7} M (pH^s ∼ 7), as shown in Figures 5b and S4c. In addition, to maintain electrical neutrality (eq 5), the concentration of perchlorate anions (ClO_4^-) is nearly zero at the PCL surface. Consequently, the IR drop within the DL becomes substantial, maintaining ϕ^s_{DL} at −1.30 V even though U is less than approximately −2.7 V (Figure S8, black curve). However, when Na^+ is present, the concentration of OH^- ions in the PCL remains low (less than 1×10^{-7} M) at $\phi^s_{DL} > -1.21$ V and becomes greater than 1×10^{-7} M (i.e., the pH^s is greater than 7) at $\phi^s_{DL} < -1.21$ V, as shown in Figure 5b. As demonstrated in Figure 4b-2, the contribution of $N_{OH,mig}$ is negligible at $\phi^s_{DL} > -1.06$ V, as indicated by the second term on the right side of eq 3. These results regarding fluxes of H^+ and OH^- ions align with trends observed in the results obtained using the planar model (Figure S9). Thus, the discussion for the porous electrode also applies to the surface of the planar electrode.

4.2. Effects of Na^+ on pH^s. As demonstrated above, the presence of Na^+ substantially affects N_H and N_{OH} at the PCL surface. How would this effect influence pH^s, which plays an important role in determining the reaction selectivity at the PCL? Figure 5a−c show plots of pH^s against U, ϕ^s_{DL}, and j_{HER}, respectively. In the absence of Na^+, pH^s asymptotically approaches 7 (Figure 5a, black curve), which is attributable to the principle of required electroneutrality (eq 5). That is, because no positively charged ions other than H^+ are present, the concentration of H^+ ions compensating for the negative charge of OH^- cannot, in principle, be less than 1×10^{-7} M (eq 4).

However, when Na^+ is present in the electrolyte, the concentration of H^+ ions can decrease to less than 1×10^{-7} M and the pH^s can increase to greater than 7 because Na^+, in addition to H^+, contributes to maintaining electroneutrality at the PCL surface. As the concentration of Na^+ increases, the U at which pH^s increases to greater than 7 shifts to a more positive value (Figure 5a), which can be explained by considering the ohmic drop. As depicted in Figure 5b, the change in pH^s with respect to ϕ^s_{DL} appears to be almost independent of the Na^+ concentration. Therefore, the increase in Na^+ concentration at the bulk electrolyte is likely to reduce the solution resistance, R_Ω, which eventually decreases U ($=\phi_{DL} + Aj_{HER}R_\Omega$ (at current collector), where the second term is the ohmic drop). As shown in Figure 5c, the pH^s begins

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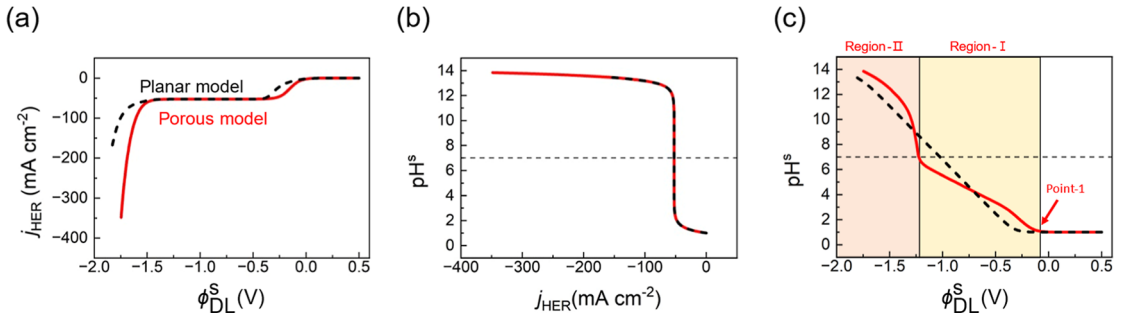


Figure 6. Comparison of simulation results obtained using planar and porous models: (a) j_{HER} vs ϕ^s_{DL}, (b) pH^s vs j_{HER}, and (c) pH^s vs ϕ^s_{DL}. The Na^+ concentration was set to 0.1 M in both models.

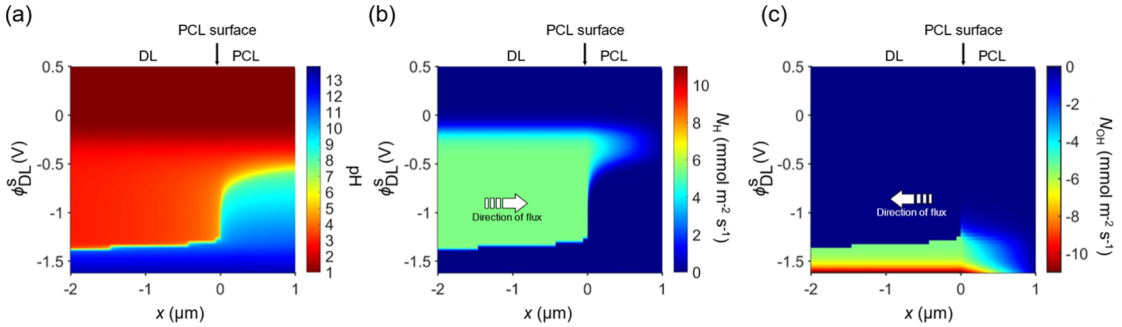


Figure 7. Contour plot of (a) pH, (b) N_H, and (c) N_{OH} values across the x- ϕ^s_{DL} plane.

increasing at lower $|j_{HER}|$ values (i.e., at a lower H^+ consumption rate) because the increase in the Na^+ concentration reduces N_H (Figure 4a-1), leading to a decrease in c_H at the PCL surface. Interestingly, trends substantially different from those in Figure 5a,b are observed when the planar model is applied (Figure S10), which will be discussed in the next section.

4.3. Influence of PCL Structure on Local pH. To further investigate the characteristics of porous electrodes, we analyzed the effect of d_{PCL} on pH^s. In this analysis, we compared the results obtained using the porous model with those obtained using the planar model. Figures 6a and S11 display the j_{HER} vs ϕ^s_{DL} and j_{HER} vs U curves, respectively, for both the planar model (illustrated with black dotted lines) and the porous model (illustrated with red lines), using a 0.1 M $HClO_4$ electrolyte containing 0.1 M $NaClO_4$. These figures indicate that the onset potentials for the reduction of H^+ and water are higher (more positive) when using the porous model. The larger electrochemical surface area (ECSA) of the porous model than that of the planar model explains the observed differences in the onset potentials. In the planar model, the ECSA is equivalent to its geometrical surface area. By contrast, the porous model exhibits a substantially larger ECSA (i.e., the specific active surface area, A_v, as defined in eq 9) because of its porous structure. This difference results in a higher value of $|j_{HER}|$ at a higher ϕ^s_{DL} for the porous model. Thus, as illustrated in Figure 6a, the j_{HER} vs ϕ^s_{DL} curve shows the expected outcomes, demonstrating the effect of having a porous layer. This comparison underscores one of the distinct characteristics of the planar and porous electrodes.

For further investigation, we analyzed the dependence of pH^s on j_{HER} and ϕ^s_{DL}. Figure 6b shows a pH^s vs j_{HER} plot for both planar and porous models. This plot leads to the unexpected conclusion that the pH^s values for the models are the same at a given j_{HER}. However, as shown in Figure 6c, the pH^s vs ϕ^s_{DL} plot displays different behavior for the planar and porous models. In the planar model (black dotted curve), the pH^s uniformly increases with a negative shift in ϕ^s_{DL} in the potential region $\phi^s_{DL} < -0.3$ V. In contrast, in the porous model (red curve), the slope of the pH^s ($dpH^s/d\phi^s_{DL}$) exhibits a complex pattern. The increase in pH^s with the negative shift of ϕ^s_{DL} starts at a higher ϕ^s_{DL} (point-1 in Figure 6c) in the porous model than in the planar model. This difference is attributed to the distinct onset potentials for H^+ reduction for the planar and porous models (Figure 6a). In the ϕ^s_{DL} range from −0.01 to −1.21 V (region-I), the $|dpH^s/d\phi^s_{DL}|$ values are lower for the porous model than that for the planar model. For Figure 6b, it means that the increase in $|j_{HER}|$ is less pronounced in the porous model than in the planar model. In the region $\phi^s_{DL} < -1.21$ V (region-II), the trends are reversed and the porous model exhibits a larger increase in pH^s as ϕ^s_{DL} shifts negatively. As a result, the porous model exhibits a higher pH^s in the potential region where ϕ^s_{DL} is lower (i.e., $\phi^s_{DL} < -1.26$ V).

The reason for the change in the $|dpH^s/d\phi^s_{DL}|$ value at pH^s ∼ 7 in the porous model can be elucidated by referring to Figure 7. Figure 7a displays a heatmap illustrating the relationships among pH, ϕ^s_{DL}, and x. The figure confirms that the local pH inside the PCL (i.e., $x > 0$) is always less than 7 when pH^s < 7 (i.e., when $\phi^s_{DL} > -1.21$ V at the PCL surface, region-I). Because the HER also occurs inside the PCL, the local current density at the PCL surface in the porous model is lower than that in the planar model under the same j_{HER} conditions. That is, the H^+ consumption rate at the PCL surface in the porous model is slower than in the planar model, which results in a slower increase in pH^s compared with that it the planar model.

In the region where pH^s > 7 and $\phi^s_{DL} < -1.21$ V (region-II), the slope of pH^s with respect to ϕ^s_{DL} ($|dpH^s/d\phi^s_{DL}|$) is larger than in the planar model (Figure 6c). This observation can be

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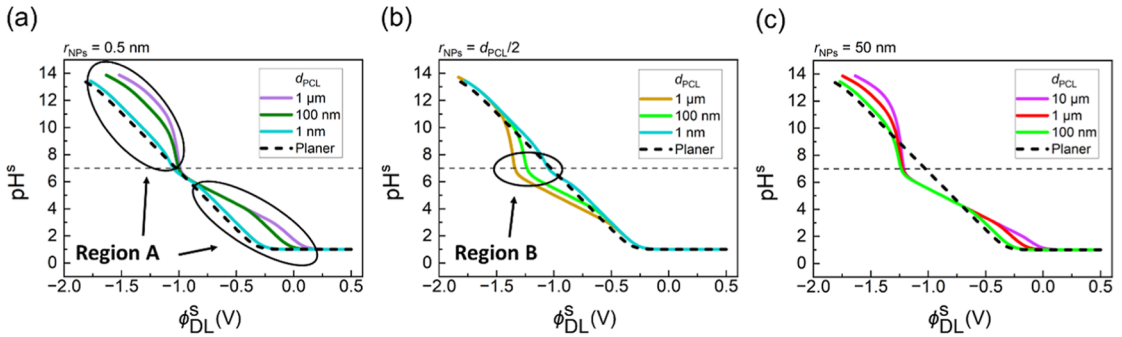


Figure 8. Changes in dependence of pH^s on $\phi^s_{DL}$when varying d_{PCL} and r_{NPs}. (a) r_{NPs} = 0.5 nm, (b) r_{NPs} = d_{PCL}/2, and (c) r_{NPs} = 50 nm (same parameter used in calculation results shown in Figures 3−7). The black dotted and the red curves in (c) are the same as those in Figure 6c.

elucidated by examining the relationships among N_H (or N_{OH}), ϕ^s_{DL}, and x (Figures 7b,c). In these figures, the specific portions of the heatmap for ϕ^s_{DL} vs x are magnified for clarity (with a comprehensive view provided in Figure S12). Figure 7b reveals that the flux of N_H from the solution bulk toward the PCL surface remains relatively constant in the range from -1.12 V < ϕ^s_{DL} < -0.40 V. The value of N_H at the PCL surface is approximately 5.4 mmol m^{-2} s^{-1} in the range from -1.12 V < ϕ^s_{DL} < -0.40 V, as shown in Figure 4a-1 (red curve). However, as shown in Figures 7c and 4b-1, the flux of OH^- (N_{OH}) transferred from the interior of the PCL toward its surface increases as ϕ^s_{DL} decreases from -1.16 V and the value of N_{OH} becomes equivalent to that of N_H at ϕ^s_{DL} = -1.21 V. Thus, the increase in the concentration of OH^- ions at the PCL surface is promoted by the increase in N_{OH} moving from the interior of the PCL toward the PCL surface, leading to a more rapid increase in pH^s. Consequently,we conclude that, in the region where the pH^s exceeds 7 (i.e., ϕ^s_{DL} < -1.21 V at the PCL surface), $|dpH^s/d\phi^s_{DL}|$ becomes larger in the porous model than in the planar model.

4.4. Dependence of Porous Model Parameters. As described above, we observed a difference in pH^s between the planar and porous models. For further analysis of how the structural parameters for a porous electrode affect the pH^s, we varied d_{PCL} and r_{NPs}. As shown in Figure S13, with the increase in d_{PCL}, the entire j_{HER} vs ϕ^s_{DL} curve shifts positively due to the increase in the effective electrode surface area. In Figure 8a, r_{NPs} was fixed at 0.5 nm, and d_{PCL}s of 1 nm, 100 nm, and 1 μm were used for the simulation. Here, the PCL contains only one layer of catalyst particles when d_{PCL} is 1 nm because the diameter of the PtNPs is equal to d_{PCL}. Then, in Figure 8b, r_{NPs} was set to half of d_{PCL}, and d_{PCL} was changed as 1, 10, and 100 nm to investigate the effect of varying r_{NPs}. Notably, changing the value of r_{NPs} will qualitatively alter the solid volume fraction inside the PCL (ε_s). However, in practice, this only slightly changes the effective diffusion coefficient according to the Bruggeman relation and does not significantly affect the phenomena of interest in this study. Therefore, we approximated ε_s as a constant value. In Figure 8c, r_{NPs} was fixed at 50 nm (the value used in other figures except for Figure 8a,b) and the effect of changing d_{PCL} was examined. For convenience, the potential range of ϕ^s_{DL} is divided into regions with high and low pH^s (region A in Figure 8a) and a region with a pH^s of approximately 7 (region B in Figure 8b).

We elucidated the behavior within region A by examining Figure 8a, which depicts how the pH^s varies with ϕ^s_{DL}. With the d_{PCL} set at 1 nm (pale blue curve), the pH^s vs ϕ^s_{DL} plot shows results almost equivalent to those for the planar model (black dotted curve). However, as d_{PCL} increases, the ϕ^s_{DL} value at which the pH^s begins to increase shifts positively and the pH^s is larger in region A compared with the corresponding pH^s predicted by the planar model. Figure 8b illustrates the outcomes when r_{NPs} was set to half the d_{PCL}. In this case, no significant change is observed in Region A; however, in Region B, the deviation from the planar model increases with increasing d_{PCL} and r_{NPs}. As shown in Figure 8c, when the d_{PCL} is varied from 100 nm to 10 μm while keeping the r_{NPs} at 50 nm, changes are observed only in region A. These results demonstrate that the d_{PCL} primarily influences the pH^s in region A, whereas the particle size strongly affects the pH^s in region B. These findings suggest that, by fine-tuning the parameters defining porous electrodes, such as d_{PCL} and the r_{NPs}, variations in local pH levels can be either enhanced or suppressed.

Finally, we present examples of how the structural parameters of a porous electrode affect the current density distribution within the pores. Figures S14a,b show the distribution of local current density and local pH within the porous electrode at various potentials. As the potential shifts negatively, the local pH within and near the porous cathode increases along with the increase in cathodic current. In this scenario, the local pH inside the porous structure tends to increase more significantly as the supply of H^+ from the solution becomes limited. When the internal pH of the porous structure exceeds 7, but the pH^s remains below 7, H^+ reduction predominantly occurs at the PCL surface, leading to higher local current density on the PCL surface compared to the interior. While the magnitude of this effect can vary depending on the numerical parameters used in the simulation, this qualitative behavior is expected to be universally observed.

5. CONCLUSIONS

In this study, we developed a simulation model that incorporates water self-ionization, enabling it to predict steady-state pH levels across a wide range (1 < pH < 14) on both the surface and inside the PCL under conditions using acidic bulk electrolytes. The inclusion of alkali-metal cations (e.g., Na^+) alongside H^+ in the model was critical. At high current densities, the absence of Na^+ leads to the pH asymptotically approaching 7, whereas its presence results in pH levels exceeding 7. Notably, while the planar model showed a linear increase in pH^s with a negative shift of ϕ^s_{DL}, the porous model displayed a more complex pattern of pH^s variation as a function of ϕ^s_{DL}. This distinction was due in part to the ECSA, which influenced the HER onset potentials for each model. Interestingly, although both models produce identical pH^s vs

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j_{HER} curves, a substantial divergence is observed in the pH^s vs ϕ^s_{DL} curves, highlighting the influence of H^+ (or OH^-) present within the PCL, which is a distinct feature of porous electrodes. In addition, this theoretical rationale would extend to analyses involving oxidation reactions in alkaline electrolytes, enabling exploration of pH decreases below 7 near the electrode. This adaptability demonstrates the model's ability to analyze reactions under various electrolyte conditions, providing insight into the complex interactions at the electrode surfaces. Thus, the simulation model developed in this study is applicable to a wide range of electrolysis systems beyond the HER. We anticipate that the model developed in this study will serve as a platform for future advanced simulation and enhance our understanding of the local pH within porous media, a critical area that remains elusive to experimental measurement.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acsami.4c09328.

LSV using Au wire as the working electrode with variable Na^+ concentration of 0.1 M $HClO_4$ electrolyte, LSV dependence on d_{DL}, schematic of the planar electrode, ion distribution near the porous electrode, effect of the effective diffusion constant on the pH distribution, flux analysis with coarser tolerance and/or mesh, LSV using the planar model, flux analysis using the planar model, ϕ^s_{DL} vs U plot using the porous model, unmagnified contour plot of N_H and N_{OH}, LSV plot dependence on d_{PCL}, and list of parameters for the simulation (PDF)

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Notes

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■ NOMENCLATURE

A	cross-sectional area (m^2)
A_v	specific active surface area of PCL (m^{-1})
c_i	concentration of species i (mol L^{-1})
d_{Bulk}	length of the bulk electrolyte (m)
d_{DL}	length of the diffusion layer of electrolyte (m)
d_{PCL}	length of the porous catalyst layer (m)
D_i	diffusion coefficient of species i in electrolyte (m^2 s^{-1})
$D_{i,eff}$	effective diffusion constant of species i inside the porous electrode (m^2 s^{-1})
$E_{HER,0}$	standard electrode potential for HER (V)
F	Faraday's constant (96485 C mol^{-1})
$j_{HER,0}$	exchange current density of HER (A m^{-2})
$j_{HER,loc}$	current density of HER at the catalyst surface of PCL (A m^{-2})
K_w	equilibrium constant of water (mol^2 L^{-2})
n_e	number of electron transfer of HER
N_i	flux of the chemical species i (mol m^{-2} s^{-1})
R	gas constant (J mol^{-1} K^{-1})
R_Ω	solution resistance (Ω)
r_{NPs}	radius of the Pt nanoparticles (m)
T	operating temperature (K)
U	applied potential (V)
$u_{m,i}$	mobility of chemical species i (m^2 V^{-1} s^{-1})
z_i	charge number of species i
α_{HER}	transfer coefficient of HER
ε_l	volume fraction of liquid phase in the PCL
ε_s	volume fraction of solid phase in the PCL
η_{HER}	overpotential of HER (V)
σ_s	electrical conductivity of nonporous composite electrode (S m^{-1})
$\sigma_{s,eff}$	effective electrical conductivity of porous catalyst (S m^{-1})
ϕ	potential (V)
ϕ_{DL}	double layer potential (V)
ϕ^s_{DL}	double layer potential at the PCL surface (V)
ϕ_l	potential in the liquid phase (V)
ϕ_s	potential in the solid phase (V)

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